

$$\begin{aligned}
 213. \text{ Given exp.} &= \frac{1}{3} \left(1 - \frac{1}{4} \right) + \frac{1}{3} \left(\frac{1}{4} - \frac{1}{7} \right) + \frac{1}{3} \left(\frac{1}{7} - \frac{1}{10} \right) \\
 &+ \frac{1}{3} \left(\frac{1}{10} - \frac{1}{13} \right) + \frac{1}{3} \left(\frac{1}{13} - \frac{1}{16} \right) \\
 &= \frac{1}{3} \left[\left(1 - \frac{1}{4} \right) + \left(\frac{1}{4} - \frac{1}{7} \right) + \left(\frac{1}{7} - \frac{1}{10} \right) \right. \\
 &\quad \left. + \left(\frac{1}{10} - \frac{1}{13} \right) + \left(\frac{1}{13} - \frac{1}{16} \right) \right] \\
 &= \frac{1}{3} \left(1 - \frac{1}{16} \right) = \frac{1}{3} \times \frac{15}{16} = \frac{5}{16}.
 \end{aligned}$$

$$\begin{aligned}
 214. \text{ Given exp.} &= \left(1 - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \left(\frac{1}{4} - \frac{1}{5} \right) \\
 &+ \dots + \left(\frac{1}{n} - \frac{1}{n+1} \right) \\
 &= \left(1 - \frac{1}{n+1} \right) = \frac{n}{n+1}.
 \end{aligned}$$

$$215. \text{ Given exp.} = \frac{4 \times 3^3 + 3^2 + 3 + 1}{4 \times 3^3} = \frac{108 + 9 + 3 + 1}{108} = \frac{121}{108}.$$

$$\begin{aligned}
 216. \text{ Given exp.} &= \frac{4 \cdot 5 \cdot 6 + 5 \cdot 6 + 2 \cdot 6 + 2 \cdot 3}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6} = \frac{120 + 30 + 12 + 6}{720} \\
 &= \frac{168}{720} = \frac{7}{30}.
 \end{aligned}$$

$$\begin{aligned}
 217. \text{ Given exp.} &= \left(\frac{1}{1^2} - \frac{1}{2^2} \right) + \left(\frac{1}{2^2} - \frac{1}{3^2} \right) + \left(\frac{1}{3^2} - \frac{1}{4^2} \right) \\
 &+ \left(\frac{1}{4^2} - \frac{1}{5^2} \right) + \dots + \left(\frac{1}{9^2} - \frac{1}{10^2} \right) \\
 &= \left(\frac{1}{1^2} - \frac{1}{10^2} \right) = \left(1 - \frac{1}{100} \right) = \frac{99}{100}.
 \end{aligned}$$

$$\begin{aligned}
 218. \text{ Given exp.} &= \frac{4-1}{4 \times 1} + \frac{9-4}{9 \times 4} + \frac{16-9}{16 \times 9} + \dots \\
 &= \left(1 - \frac{1}{4} \right) + \left(\frac{1}{4} - \frac{1}{9} \right) + \left(\frac{1}{9} - \frac{1}{16} \right) + \dots \\
 &= \left(\frac{1}{1^2} - \frac{1}{2^2} \right) + \left(\frac{1}{2^2} - \frac{1}{3^2} \right) + \left(\frac{1}{3^2} - \frac{1}{4^2} \right) + \dots
 \end{aligned}$$

$$\therefore 99\text{th term of the series} = \left(\frac{1}{99^2} - \frac{1}{100^2} \right).$$

$$\begin{aligned}
 \therefore \text{ Given exp.} &= \left(\frac{1}{1^2} - \frac{1}{2^2} \right) + \left(\frac{1}{2^2} - \frac{1}{3^2} \right) + \left(\frac{1}{3^2} - \frac{1}{4^2} \right) \\
 &+ \dots + \left(\frac{1}{98^2} - \frac{1}{99^2} \right) + \left(\frac{1}{99^2} - \frac{1}{100^2} \right) \\
 &= \left(1 - \frac{1}{100^2} \right) = \left(1 - \frac{1}{10000} \right) = \frac{9999}{10000}.
 \end{aligned}$$

219. Suppose Mayank, Mirza and Little paid ₹ x , ₹ y and ₹ z respectively.

$$\text{Then, } x = \frac{1}{2} (60000 - x), y = \frac{1}{3} (60000 - y), z = \frac{1}{4} (60000 - z).$$

$$\Rightarrow 2x = 60000 - x, 3y = 60000 - y, 4z = 60000 - z$$

$$\Rightarrow 3x = 60000, 4y = 60000, 5z = 60000$$

$$\Rightarrow x = 20000, y = 15000, z = 12000.$$

Amount paid by Jaspal

$$= ₹ [60000 - (20000 + 15000 + 12000)] = ₹ 13000.$$

220. Income after 1 year = ₹ (4×2^1) lakhs.

Income after 2 years = ₹ $(4 \times 2 \times 2)$ lakhs

$$= ₹ (4 \times 2^2) \text{ lakhs}$$

$\therefore$ Income after 5 years = ₹ (4×2^5) lakhs = ₹ 128 lakhs

$$= ₹ 1.28 \text{ crores.}$$

221. Since the population doubles every day, so it was half-full on 27/2/2007 and quarter-full on 26/2/2007.

222. Since the number doubles every minute, so the basket was $\frac{1}{2^n}$ full in $(60 - n)$ minutes.

Thus, it was $\frac{1}{32} \left(= \frac{1}{2^5} \right)$ full in $(60 - 5)$ i.e., 55 minutes.

223. Total amount received by the man in 2010

$$= ₹ [(5000 \times 2^4) + (5000 \times 2^3) + (5000 \times 2^2) + (5000 \times 2)]$$

$$= ₹ (80000 + 40000 + 20000 + 10000) = ₹ 150000.$$

224. True length of the rope = 120 m + (3×120) cm

$$= 120 \text{ m} + 360 \text{ cm}$$

$$= 120 \text{ m} + 3 \text{ m } 60 \text{ cm} = 123 \text{ m } 60 \text{ cm.}$$

225. Let the number be x . Then, $63x - 36x = 3834$

$$\Leftrightarrow 27x = 3834$$

$$\Leftrightarrow x = 142.$$

226. Among the given numbers, only 60489 is a multiple of 423.

227. Let the given number be x .

$$\text{Then, } \frac{7}{8}x - \frac{7}{18}x = 770 \Leftrightarrow \frac{63x - 28x}{72} = 770 \Leftrightarrow \frac{35x}{72} = 770$$

$$\Leftrightarrow x = \left(\frac{770 \times 72}{35} \right) = 1584.$$

228. Let the number be x .

$$\text{Then, } \frac{17}{8}x - \frac{8}{17}x = 225 \Leftrightarrow \frac{289x - 64x}{136} = 225$$

$$\Leftrightarrow \frac{225}{136}x = 225 \Leftrightarrow x = 136.$$

$$\therefore \text{ Expected answer} = 136 + \frac{17}{8} = \left(136 \times \frac{8}{17} \right) = 64.$$

229. Let the number be $2x$.

$$\text{Then, } \left(\frac{x}{6} + \frac{x}{4} \right) - \frac{2x}{5} = 4 \Leftrightarrow \frac{10x + 15x - 24x}{60} = 4 \Leftrightarrow x = 240.$$

Hence, required number = $2x = 480$.

230. Final answer = $\left(9\frac{1}{9} - 3\frac{2}{3} \right) \times 450 = \left(\frac{82}{9} - \frac{11}{3} \right) \times 450$

$$= \left(\frac{49}{9} \times 450 \right) = 2450.$$

231. $\frac{7}{8}$ of 1008 - $\frac{3}{4}$ of 968 = $\left(\frac{7}{8} \times 1008 \right) - \left(\frac{3}{4} \times 968 \right)$

$$= 882 - 726 = 156.$$

232. $x = y - \frac{y}{10} \Rightarrow x = \frac{9y}{10} \Rightarrow x = \frac{9}{10} \times y \Rightarrow x \propto y$.

Thus, as y increases, x also increases.